Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\mathcal{L}=\frac{1}{2}\left\{\partial_m\phi_i\partial^m\phi_i+\imath\bar{\psi}_i\gamma^m\partial_m\psi_i-\frac{\partial\mathcal{W}}{\partial\phi_i}\frac{\partial\mathcal{W}}{\partial\phi_i}-\frac{\partial^2\mathcal{W}}{\partial\phi_i\partial\phi_j}\bar{\psi}_i\psi_j\right\}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\Delta H^{\epsilon}=d(d\widetilde{d}_{\epsilon}\times\widetilde{d}_{\epsilon})=d\widetilde{d}_{\epsilon}\times d\widetilde{d}_{\epsilon}=\sum_{i,j=1}^{3}\frac{\partial\widetilde{d}_{\epsilon}}{\partial x_i}\times\frac{\partial\widetilde{d}_{\epsilon}}{\partial x_j}dx_i\wedge dx_j\text{ in }\mathbb{R}^3.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\partial_{\varphi}\left\{\partial_z S^z\partial_{\bar{z}}S^{\bar{z}}-\partial_z S^{\bar{z}}\partial_{\bar{z}}S^z+\mathcal{H}\right\}=\partial_z p+\partial_{\bar{z}}\bar{p}+\partial_{\varphi}\mathcal{H}=\partial_{\varphi}\left\{\partial_z S+\partial_{\bar{z}}\bar{S}+\mathcal{H}\right\}=0\,.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$I_{\psi,\widetilde{\psi}}\equiv\frac{1}{2}\int_M\epsilon^{\mu\nu\rho\sigma}\left[\widetilde{\psi}^{A'}{}_{\mu}\,e_{AA'\nu}\,D_{\rho}\,\psi^{A}{}_{\sigma}-\psi^{A}{}_{\mu}\,e_{AA'\nu}\,D_{\rho}\,\widetilde{\psi}^{A'}{}_{\sigma}\right]d^4x\ ,$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\Psi_1\left(\boldsymbol{d}_{\mathcal{S}_1}(s_1,s_1')^p,\rho_2(y_1,y_1')^p\right)=\Psi_1\circ\psi_1\left(\boldsymbol{d}_{\mathcal{Y}_1}(y_1,y_1')^p,\boldsymbol{d}_{\mathcal{Y}_2}(y_2,y_2')^p,\boldsymbol{d}_{\mathcal{X}}(x,x')^p\right),$$